

Final Project

Design Document

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Introduction

Project Functionality

The play area consists of tiles, which are 24 pixels in size. The play area is 32 tiles wide (768 pixels) and 24 tiles tall (576 pixels). The snake starts with a length of 1 tile. It moves in one of four directions (Up, Down, Left, Right) until the player changes direction using WASD or arrow keys. A food item will spawn on a random tile and will never spawn on top of the snake. The food item will respawn at a random tile after the snake's head touches it. In addition, 1 point will be awarded to the player for each food item eaten and the tail will grow by 1 tile. Score is displayed in the top left corner of the play area. The game will end if the head goes outside of the play area or if the snake collides with itself. When the game ends, a game over screen will appear and display the total score. The player can then restart the game by pressing any key or quit. Hunger was never implemented but still remains in the code.

Design Process

The design process consisted of the following:

1. A rough framework was made to establish the core gameplay mechanics. Pseudocode and a flowchart were created to set a clear example of what the code should be like.
2. The code was refactored and tested. The code was rewritten from the ground up with an object-oriented design in mind. For example, a class was created for the snake. A snake object is created in the main loop. Its methods and attributes are easily accessible and organized.
3. The final step was to create an art style. The background has a checker pattern. The snake and fruit have different colors. The game over screen has text instead of a single print statement.

Project Development

Pseudocode

START GAME TILE_SIZE = 64

BOARD_SIZE = 16

FRAME_RATE = 60

Construct game board (TILE_SIZE, BOARD_SIZE)

length = 3

direction = RIGHT

Spawn food ()

hunger_timer = 4

score = 0

LOOP WHILE game is running: IF user input (WASD or Arrow Keys): Change direction

Calculate head = snake_head + direction

IF head is outside board OR inside snake body:

 END GAME

IF head is at food position:

 Increase score by 1

 Reset hunger_timer

 Spawn food ()

ELSE:

 Subtract 1 from hunger_timer

IF hunger_timer == 0:

 IF snake length > 1:

 Remove tail segment

 Reset hunger_timer

 ELSE:

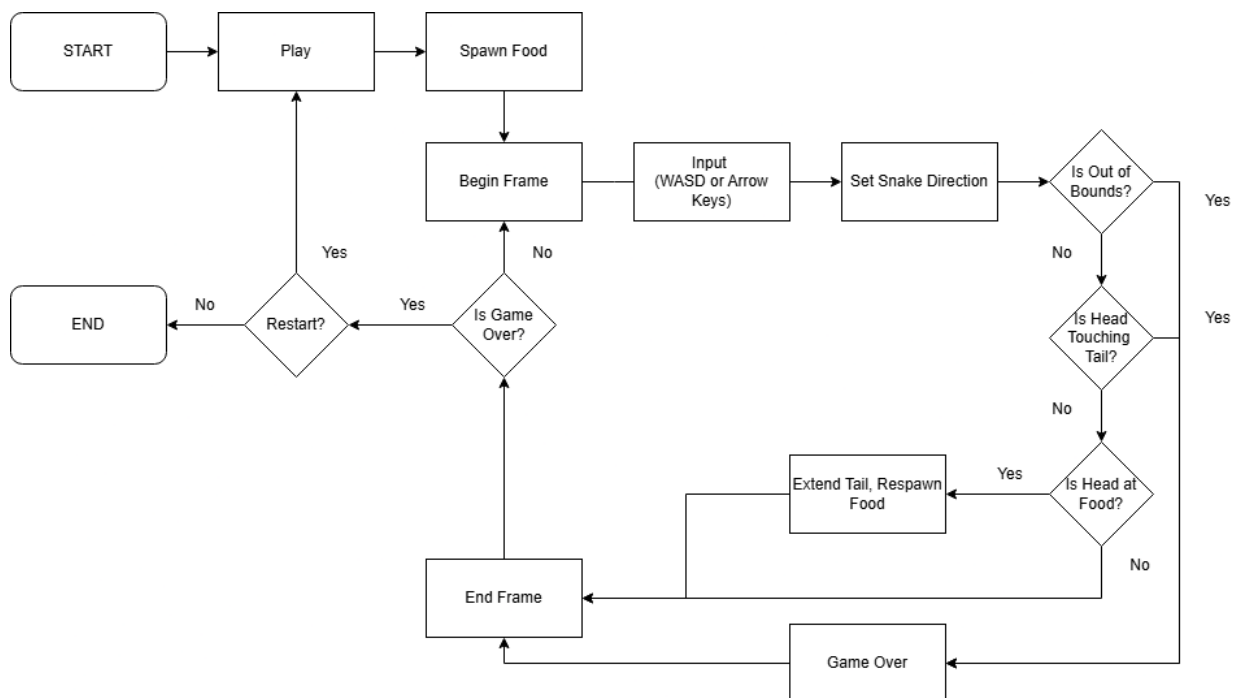
END GAME

Wait for next frame

END LOOP

This is a draft of the main code for the project, specifically the core game loop or algorithm. When the game starts the board is constructed and variables are assigned. The plan is that the code will loop every frame using functions that have already been defined. I want to use classes or some sort of object-oriented design.

Flowchart



This is the flowchart made with draw.io (<https://app.diagrams.net/>) for the game without the abandoned hunger mechanic.

Requirements

1. Board Size / Play Area: The game should be played in a window containing at least a 10-by-10 tile set.

The game is at least a 10-by-10 tile set and is completely customizable.

2. Snake Movement: The snake should move continuously in one of four directions (up, down, left, right). The player should control the direction using the arrow keys.
The snake can move in all four directions continuously. The direction can be controlled by arrow keys or WASD.
3. Snake Growth: Every time the snake eats a food item, it should grow longer by one unit.
When the snake touches a food item, the tail grows by one unit.
4. Food Generation: Food should appear randomly on the game screen after being eaten by the snake.
Food items will respawn at a random tile after being eaten by the snake. They will never spawn on top of a snake.
5. Collision Detection: The game should end if the snake collides with the walls of the screen or its own body.
The game ends if the snake goes outside the play area or collides with itself.
6. Game Over and Score: Display a Game Over message when the snake dies.
When the game ends a game over message appears. Any key can be pressed to restart the game.